

Let $n = p_1 p_2$ where $p_1, p_2 \in \text{Primes}$

$\Rightarrow$ factors of $n = 1, p_1, p_2, p_1 p_2$

Consider $1 + p_1 + p_2 + \cancel{p_1 p_2} > \cancel{2 p_1 p_2}$

$$\Rightarrow 1 + p_1 + p_2 > p_1 p_2$$

$$\Rightarrow \frac{1 + p_1 + p_2}{p_1 p_2} > 0$$